CLINIC APPOINTMENT OPERATIONS AND NO-SHOW ANALYSIS

Executive Summary | Medical Appointment No-Shows Dataset | Brazil, 2016

SECTION	EXECUTIVE FINDINGS
Project Overview	This independent analytical portfolio project examines 110,527 scheduled appointments to understand attendance patterns, missed appointments and their relationships with appointment characteristics and scheduling patterns.
Overall Performance	Of 110,527 appointments, 88,208 were attended and 22,319 were missed. The overall attendance rate was 79.81%, while the no-show rate was 20.19%.
Waiting-time Pattern	Missed appointments had a higher median waiting time (11 days) than attended appointments (2 days). No-show rates generally increased across longer waiting-time groups up to 31–60 days, before declining in the more-than-60-days group.
SMS Reminder Comparison	The no-show rate was 27.57% among appointments with a recorded SMS reminder, compared with 16.70% among appointments without a recorded SMS reminder. This represents a difference of 10.87 percentage points and does not establish a causal effect of SMS reminders.
Weekday Pattern	Wednesday recorded the highest appointment volume (25,867). Friday had the highest no-show rate among weekdays with substantial appointment volume (21.23%), while Thursday had the lowest (19.35%).
Neighbourhood Pattern	Santos Dumont recorded the highest no-show rate among the 60 neighbourhoods included in the main comparison (28.92%). Jardim Camburi recorded the highest number of missed appointments (1,465), despite a lower no-show rate of 18.98%.
Operational Implications	Waiting-time groups, weekday patterns and neighbourhood-level differences can be monitored to identify areas for further operational investigation. The SMS comparison also warrants further examination before drawing conclusions about reminder effectiveness.
Limitations and Interpretation	The findings are descriptive and based on observational data from 2016. They do not establish causation. Five records with negative calendar waiting times were excluded from waiting-time analysis only. Saturday had just 39 appointments, and the main neighbourhood comparison used a pragmatic minimum-volume threshold of 400 appointments.
Overall Conclusion	The analysis identifies differences in attendance patterns across waiting-time groups, SMS categories, weekdays and neighbourhoods. These findings provide a basis for operational monitoring and further investigation into missed appointments.